

Sensor and Systems

Trilateration-based sensors

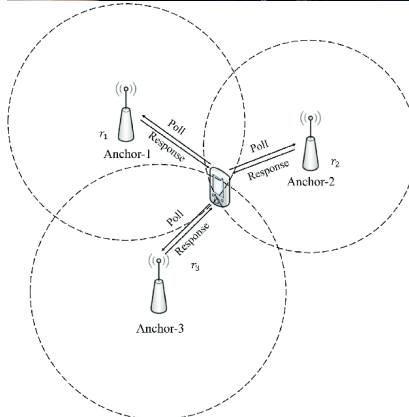
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Electronic Systems

8th semester

- ▶ Trilateration-based sensors
 - ▶ Bluetooth RSSI
 - ▶ Games-on-Track
 - ▶ Underwater sensor networks
- ▶ Global Positioning System
 - ▶ (Approximate) Geometry of the Earth
 - ▶ GPS satellites
 - ▶ GPS data transmission
- ▶ GPS trilateration
 - ▶ Pseudorange coding
 - ▶ Position estimation Error models

Trilateration-based sensors





- Distance ρ to known transmitter is estimated by free-space path loss:

$$\ell = \left(\frac{4\pi\rho}{\lambda} \right)^\gamma = \left(\frac{4\pi\rho f}{c} \right)^\gamma$$

where λ is wavelength, f is frequency, c is the speed of light, and γ is approximately 2.

- Received Signal Strength Indicator:

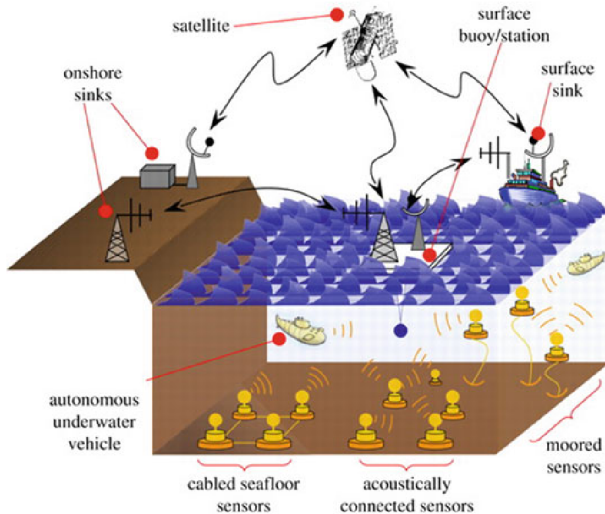
$$S = -10\gamma \log_{10} \rho + 10\gamma \log_{10} \left(\frac{4\pi f}{c} \right)$$

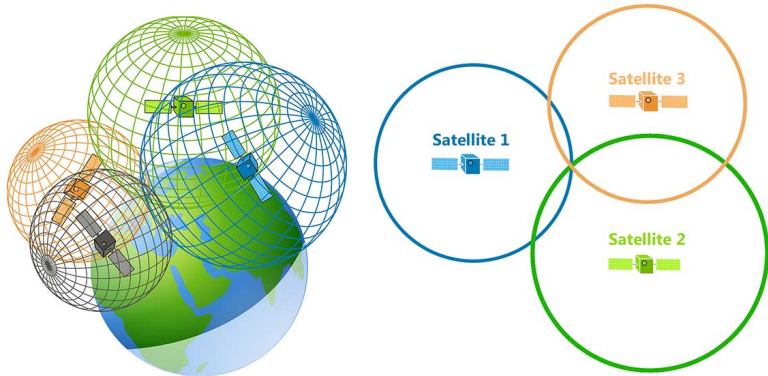


<https://www.youtube.com/@GamesOnTrack/videos>

- ▶ Uses radio 'chirps' to determine the distance to known beacons
- ▶ Used to detect and track model trains.
- ▶ Used in several student projects at AAU in the past.
- ▶ Unfortunately the company went belly-up...

Underwater sensor networks

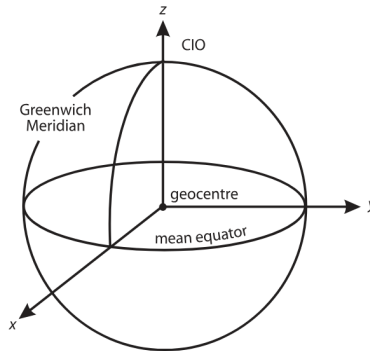




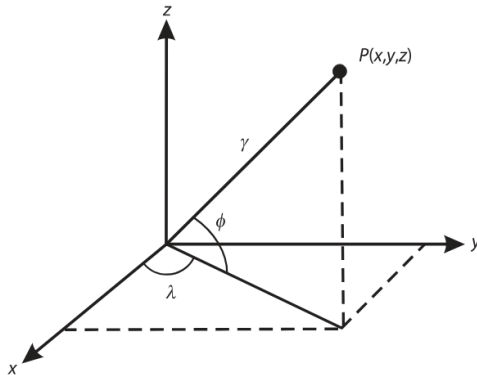
- ▶ Minimum requirement for standard trilateration: 3 ranges to 3 known points.
- ▶ GPS requirement: 4 "pseudoranges" to 4 satellites at known coordinates.

(Approximate) Geometry of the Earth

Earth-Centered Earth-Fixed coordinates



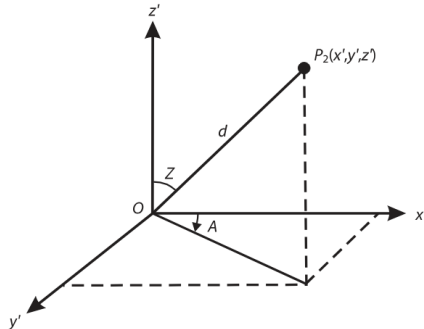
- ▶ The ECEF coordinate system is a right-handed Cartesian system with the origin at the Earth's centre of mass.
- ▶ z -axis and the mean rotational axis of the Earth coincide.
- ▶ x -axis points to the mean Greenwich meridian.
- ▶ Geographic coordinates (Longitude and Latitude, WGS84) - measured in degrees, minutes, seconds.
- ▶ One latitudinal second measures 30.715 m, one latitudinal minute is 1843 m and one latitudinal degree is 110.6 km.



- Point P in ECEF (global) coordinates:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \gamma \begin{bmatrix} \cos \phi \cos \lambda \\ \cos \phi \sin \lambda \\ \sin \phi \end{bmatrix} \quad \text{and} \quad \begin{aligned} \gamma &= \sqrt{x^2 + y^2 + z^2} \\ \tan \lambda &= y/x \\ \tan \phi &= z/\sqrt{x^2 + y^2} \end{aligned}$$

where λ is longitude, ϕ is latitude and γ is often taken as R_E , the Earth's mean radius.



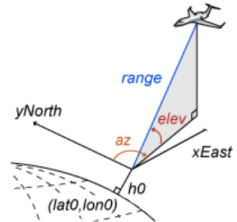
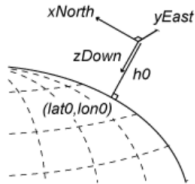
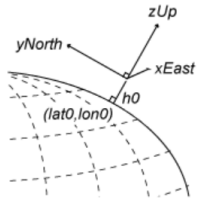
- ▶ Let $(\mathbf{x}', \mathbf{y}', \mathbf{z}')$ be a local coordinate system. The $\mathbf{x}', \mathbf{y}'$ plane¹ is normally referred to as the *horizontal plane*.
- ▶ A and Z are known as *Azimuth* and *Zenith* (angles), respectively.
- ▶ For point P_2 one has:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = d \begin{bmatrix} \cos A \sin Z \\ \sin A \sin Z \\ \cos Z \end{bmatrix} \quad \text{and} \quad \begin{aligned} d &= \sqrt{x'^2 + y'^2 + z'^2} \\ \tan A &= y'/x' \\ \cos Z &= z'/d \end{aligned}$$

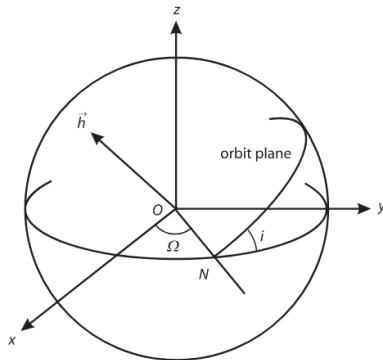
¹Obs: $\mathbf{x}'$ and $\mathbf{y}'$ should be swapped in the figure above.

(Approximate) Geometry of the Earth

Common local coordinate systems



- ▶ Left to right: North-East-Down, East-North-Down, Azimuth-Elevation-Range



- Satellite position vector r in ECEF coordinates obeys:

$$\ddot{\mathbf{r}} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = -\frac{GM}{\|\mathbf{r}\|^3} \mathbf{r}$$

- Solution: Elliptic orbit in *orbital plane*.
- The orbital plane is perpendicular to the *angular momentum* $\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}}$.

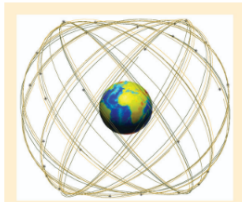


FIGURE 1 Satellite orbits as seen in Inertial frame

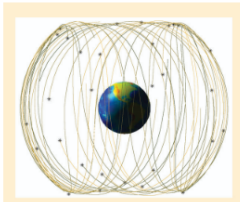


FIGURE 2 Satellite orbits as seen in Earth Centered Earth Fixed frame

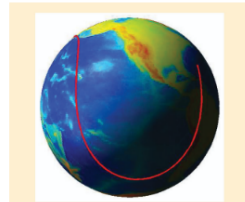


FIGURE 3 Sub-satellite points for a selected satellite

K. Borre, "GPS EasySuite II", InsideGNSS June 2010

- ▶ Satellites transmit a "Navigation Message" toward Earth.
- ▶ Includes orbit parameters (broadcast ephemeris) for coordinate calculation.
- ▶ Coordinates (x, y, z) are in the WGS-84 geocentric system.
- ▶ Satellite speed: Approximately 4 km/s relative to Earth's center.
- ▶ Relative speed along line of sight (range rate): Typical values approx. 1 km/s.
- ▶ Orbital period $T \approx 11$ hr 58 min; 2 revolutions per sidereal day.
- ▶ Result: Repeating ground tracks and sky tracks every sidereal day.

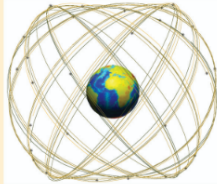


FIGURE 1 Satellite orbits as seen in Inertial frame

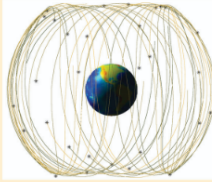


FIGURE 2 Satellite orbits as seen in Earth Centered Earth Fixed frame

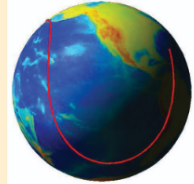
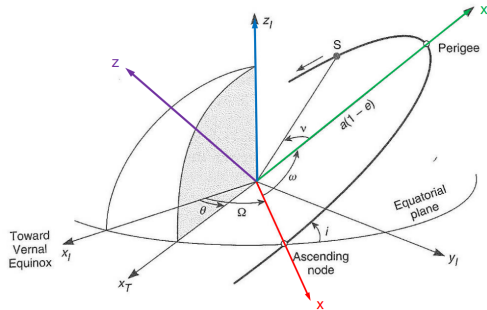


FIGURE 3 Sub-satellite points for a selected satellite

K. Borre, "GPS EasySuite II", InsideGNSS June 2010

- ▶ 24+ satellites (a.k.a. "Space Vehicles")
- ▶ 6 orbital planes
- ▶ 55 degree inclination
- ▶ Elliptic (albeit nearly circular) orbits
- ▶ Average orbital radius: $r = 26,561$ km

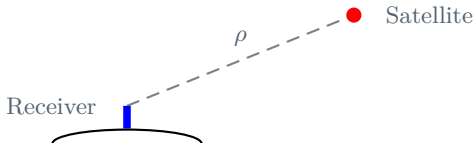


► Position in ECEF:

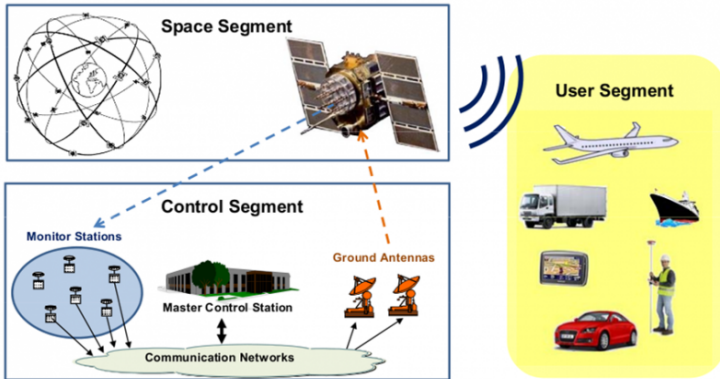
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \Omega & -\sin \Omega & 0 \\ \sin \Omega & \cos \Omega & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$$

$$= \begin{bmatrix} x' \cos \Omega - y' \sin \Omega \cos i \\ x' \sin \Omega + y' \cos \Omega \sin i \\ y' \sin i \end{bmatrix}$$

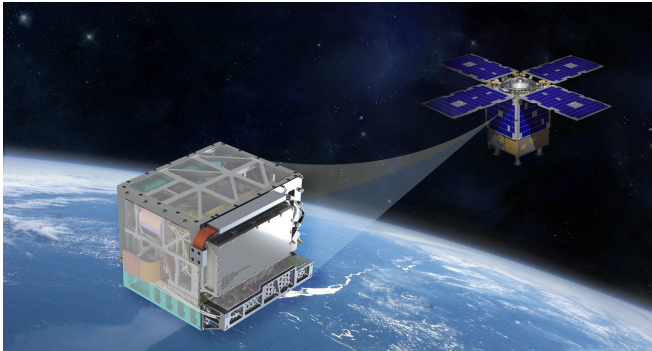
where $x' = r \cos(\nu + \omega)$, $y' = r \sin(\nu + \omega)$, and $z' = 0$.



- ▶ Signal transmission time is encoded using an onboard atomic clock.
- ▶ Reception time is recorded by the receiver's own clock.
- ▶ Time-of-Flight (pseudorange) $\rho = c\Delta t$, where Δt is time difference between transmission and reception, and c is the speed of light.
- ▶ Includes clock errors because receiver clocks are imperfect.
- ▶ Satellite clock error is provided in the Navigation Message as a bit-wise polynomial.
- ▶ Receiver clock error is an unknown and must be estimated alongside receiver coordinates, giving a total of 4 unknowns (3 coordinates + 1 receiver clock error).



- ▶ **Space Segment:** Constellation of satellites transmitting signals.
- ▶ **Control Segment:** Responsible for monitoring and operating the satellites (uploads Keplerian parameters once every hour, as well as various other maintenance data).
- ▶ **User Segment:** User hardware and software for positioning and timing.
- ▶ **Ground Segment:** Tracking networks providing reference control and services.



- ▶ Hardware: Atomic clocks (2 rubidium, 2 cesium), solar panels, Ni-Cd batteries.
- ▶ Control Segment monitoring: Assess satellite health and predict orbits/clocks.
- ▶ Maneuvers: Satellites can be moved to optimize geometry or adjust orbits.
- ▶ Clock management: Switching between available on-board clocks if needed.

GPS data output string

Examples of GPS “sentences”



```
$GPRMC,044840.00,A,3401.21044,N,11824.67722,W,0.165,,010621,,,D*60
$GPVTG,,T,,M,0.165,N,0.306,K,D*21
$GPGGA,044840.00,3401.21044,N,11824.67722,W,2,07,1.34,29.5,M,-32.9,M,
$GPGSA,A,3,21,32,46,27,08,22,10,,,,,3.07,1.34,2.76*0F
$GPGSV,3,1,12,01,18,319,17,08,27,257,20,10,44,068,33,18,00,142,*73
$GPGSV,3,2,12,21,39,313,23,22,20,296,14,23,16,089,,27,22,221,32*7A
$GPGSV,3,3,12,31,28,168,20,32,70,025,31,46,49,199,30,51,49,161,*7E
$GPGLL,3401.21044,N,11824.67722,W,044840.00,A,D*7F
```

- ▶ GPGGA: Global positioning system fix data (time, position, fix type data)
- ▶ GPGLL: Geographic position, latitude, longitude
- ▶ GPVTG: Course and speed information relative to the ground
- ▶ GPRMC: Time, date, position, course, and speed data
- ▶ GPGSA: GPS receiver operating mode, satellites used in the position solution, and DOP values.
- ▶ GPGSV: The number of GPS satellites in view, satellite ID numbers, elevation, azimuth, and SNR values.

See e.g., <https://www.redhat.com/en/blog/architects-guide-gps-and-gps-data-formats>

GPS data output string

GPGGA, Global Positioning System Fix Data



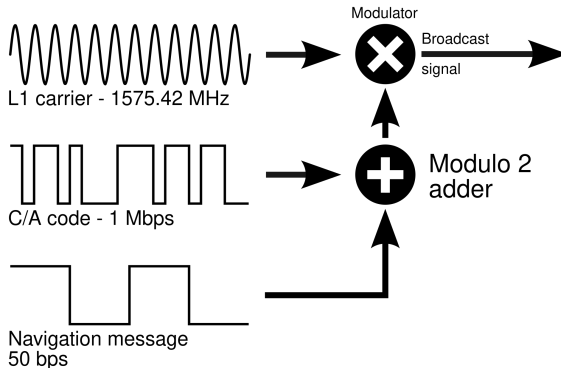
\$GPGGA,170241.00,3401.21189,N,11824.67797,W,1,06,2.14,71.6,M,-32.9,M

Field	Name	Data	Description
0	Type ID	\$GPGGA	GGA protocol header
1	Time	170241	17:02:41 UTC
2	Latitude	3401.21189	ddmm.mmmm format, converts to 34.020196 or 34d 1' 12.706" N
3	Hemisphere	N	N = North, S = South
4	Longitude	11824.67797	dddmm.mmmm format, converts to -118.41129833 or 118d 24' 48673" W
5	Hemisphere	W	W = West, E = East
6	fix type	1	Data is from a GPS fix
7	No. sats	06	6 Satellites are in view
8	HDOP	2.14	Relative accuracy of horizontal position
9	Alt.	71.6	71.6 meters above mean sea level
10	Alt. unit	M	M = meters
11	Geoid height	-32.9	-32.9 meters above WGS84 ellipsoid
12	Height unit	M	M = meters
13	Time		Time since last DGPS update (no last update)
14	DGPS ID		DGPS station id (no station ID)
15	Checksum	*57	Check for transmission errors

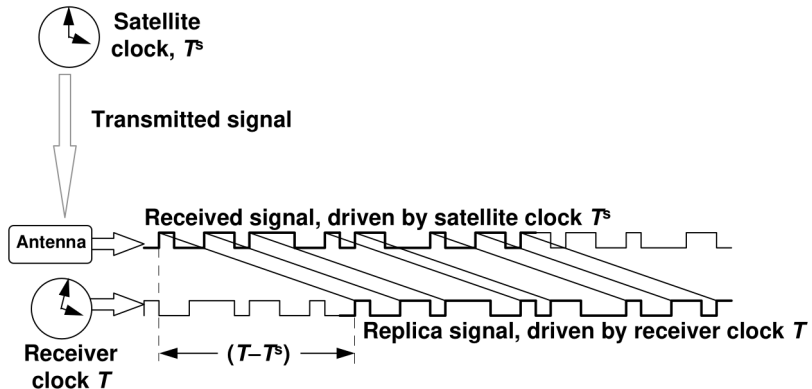
The navigation message consists of 30-second frames 1,500 bits long, divided into five 6-second subframes of ten 30-bit words each:

Sub-frame	Word	Description
1	1-2	Telemetry and handover words (TLM and HOW)
	3-10	Satellite clock, GPS time relationship
2-3	1-2	Telemetry and handover words (TLM and HOW)
	3-10	Ephemeris (precise satellite orbit)
4-5	1-2	Telemetry and handover words (TLM and HOW)
	3-10	Almanac component (satellite network synopsis, error correction)

- ▶ *Ephemeris* is precise orbital information for the transmitting satellite.
- ▶ *Almanac* consists of coarse orbit and status information for each satellite in the constellation, an ionospheric model, and information to relate GPS derived time to Coordinated Universal Time (UTC).



- ▶ Fundamental frequency: 10.23 MHz.
- ▶ L1 Channel: 1575.42 MHz (wavelength: 19.0 cm).
- ▶ L2 Channel: 1227.60 MHz (wavelength: 24.4 cm).
- ▶ Encoding: Phase modulation using binary bits (+1 or -1).



- ▶ The receiver can generate a replica of the expected satellite signal
- ▶ When the two signals are not properly matched in time, the result of autocorrelation of bitwise multiplication of signal and replica gives an answer close to zero;
- ▶ If the signals are matched in time, the result is close to +1. This gives the difference between the receiver and satellite clocks

- Observation equation for distance to satellite S :

$$P_S = (T - T_S)c$$

- Incorporating clock biases $T = t + \tau$ (receiver) and $T_S = t_S + \tau_S$ (satellite):

$$P_S = (t + \tau)c - (t_S + \tau_S) = \rho(t, t_S) + c\tau - c\tau_S$$

- Geometric distance between receiver and satellite:

$$\rho(t, t_S) = \sqrt{(x_S(t_S) - x(t))^2 + (y_S(t_S) - y(t))^2 + (z_S(t_S) - z(t))^2}$$

- $(x_S(t_S), y_S(t_S), z_S(t_S))$ and τ_S are transmitted from the satellite; that leaves 4 unknowns: $(x(t), y(t), z(t))$ and τ .
- Ignores atmospheric delays and relativity for simplicity.

- For $m \geq 4$ satellites in view:

$$P_1 = \sqrt{(x_1(t_1) - x(t))^2 + (y_1(t_1) - y(t))^2 + (z_1(t_1) - z(t))^2} + c\tau - c\tau_1$$

$$P_2 = \sqrt{(x_2(t_2) - x(t))^2 + (y_2(t_2) - y(t))^2 + (z_2(t_2) - z(t))^2} + c\tau - c\tau_2$$

$$\vdots$$

$$P_m = \sqrt{(x_m(t_m) - x(t))^2 + (y_m(t_m) - y(t))^2 + (z_m(t_m) - z(t))^2} + c\tau - c\tau_m$$

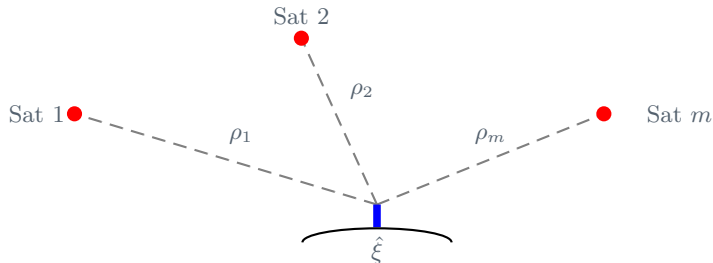
- Linearize around (x_0, y_0, z_0, τ_0) to obtain

$$b = A\hat{\xi} + \nu$$

where

$$\hat{\xi} = \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \\ \tau - \tau_0 \end{bmatrix}, \quad b = \begin{bmatrix} P_1 - P_0 \\ P_2 - P_0 \\ \vdots \\ P_m - P_0 \end{bmatrix}, \quad A = \begin{bmatrix} \frac{\partial P_1}{\partial x} & \frac{\partial P_1}{\partial y} & \frac{\partial P_1}{\partial z} & \frac{\partial P_1}{\partial \tau} \\ \frac{\partial P_2}{\partial x} & \frac{\partial P_2}{\partial y} & \frac{\partial P_2}{\partial z} & \frac{\partial P_2}{\partial \tau} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial P_m}{\partial x} & \frac{\partial P_m}{\partial y} & \frac{\partial P_m}{\partial z} & \frac{\partial P_m}{\partial \tau} \end{bmatrix}$$

and ν contains noise terms.



- ▶ Minimize the sum of squares of the estimated residuals: $J(\hat{\xi}) = \nu^\top \nu$.

- ▶ Calculated as

$$\hat{\xi} = (A^\top A)^{-1} A^\top b$$

- ▶ Assumes $A^\top A$ is non-singular, implying that satellites must not be in the same plane.
- ▶ In practice, $m \geq 5$ is usually sufficient for a robust solution.

- Mapping observation ‘noise’ ν to estimate errors ν_ξ :

$$\nu_\xi = (A^\top A)^{-1} A^\top \nu$$

- Covariance matrix Σ_ξ describes the precision of the estimates:

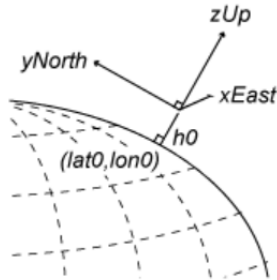
$$\Sigma_\xi = (A^\top A)^{-1} A^\top \Sigma_\nu A (A^\top A)^{-1}$$

- In particular, if $\Sigma_\nu = \sigma_\nu^2 I$:

$$\Sigma_\xi = \sigma_\nu^2 (A^\top A)^{-1}$$

where σ_ν^2 is the estimated (expected) noise variance for all the observations.

- Diagonal elements of Σ_ξ provide the variance for each coordinate and clock bias.

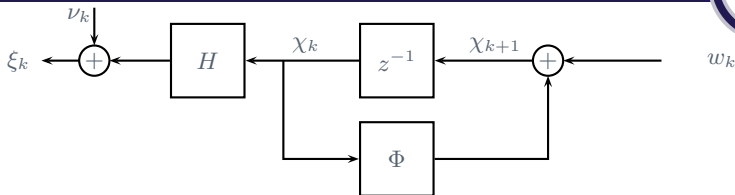


- Error estimates in ECEF may be transformed into estimates in local North-East-Up coordinates by

$$\nu_{\text{NEU}} = R_{\phi\lambda} \nu_{\xi} = \begin{bmatrix} -\sin \phi \cos \lambda & -\sin \phi \sin \lambda & \cos \phi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \phi \cos \lambda & \cos \phi \sin \lambda & \sin \phi \end{bmatrix} \nu_{\xi}$$

- Covariance matrix in local coordinates:

$$\Sigma_{\text{NEU}} = R_{\phi\lambda} \Sigma_{\xi} R_{\phi\lambda}^{\top}.$$



- Obviously, a Kalman estimator can now be used to estimate position and velocity of moving objects based on consecutive GPS observations.
- As usual, if no other information is available, the model above can be used with

$$\chi_k = \begin{bmatrix} x_k \\ y_k \\ z_k \\ v_{x,k} \\ v_{y,k} \\ v_{z,k} \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 0 & 0 & T_s & 0 & 0 \\ 0 & 1 & 0 & 0 & T_s & 0 \\ 0 & 0 & 1 & 0 & 0 & T_s \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} I & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 6}$$

and w, ν the usual Gaussian noise signals.

- In particular, the covariance of ν can be estimated as on the previous slides.

- ▶ Trilateration-based sensors use range information to estimate position
- ▶ The Global Positioning System (GPS) is a satellite-based system that has been in operation since the 1980s.
- ▶ It relies on transmission of certain key variables as well as travel time of the data packets themselves to provide range information.
- ▶ Position estimation is performed using local linearization and least squares methods.
- ▶ Errors estimated in global coordinates may be transformed to local coordinates, where they arguably make more sense.